

A. Within- and between-class diversity

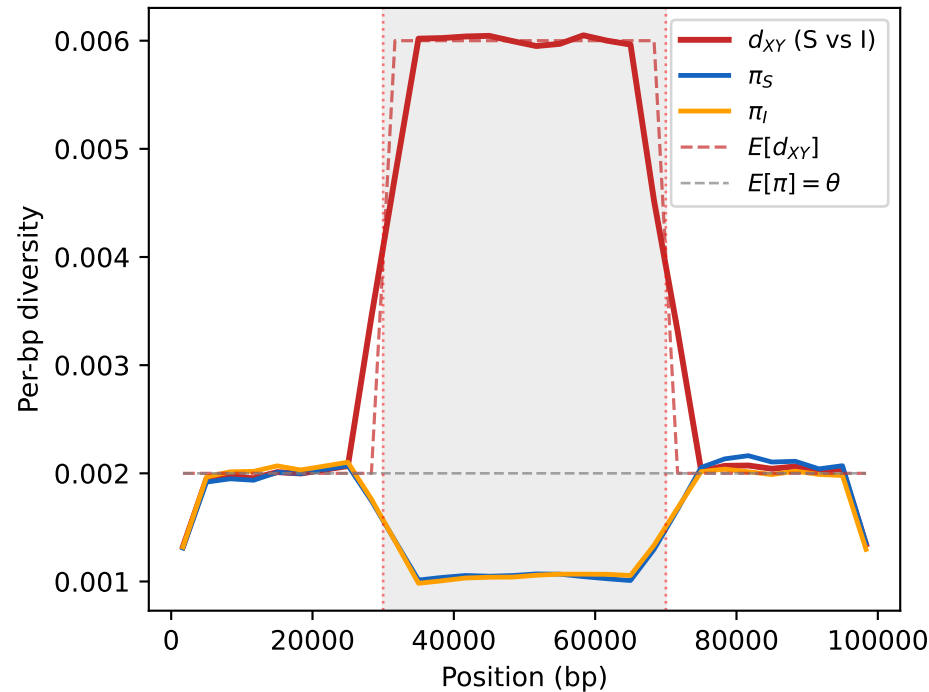
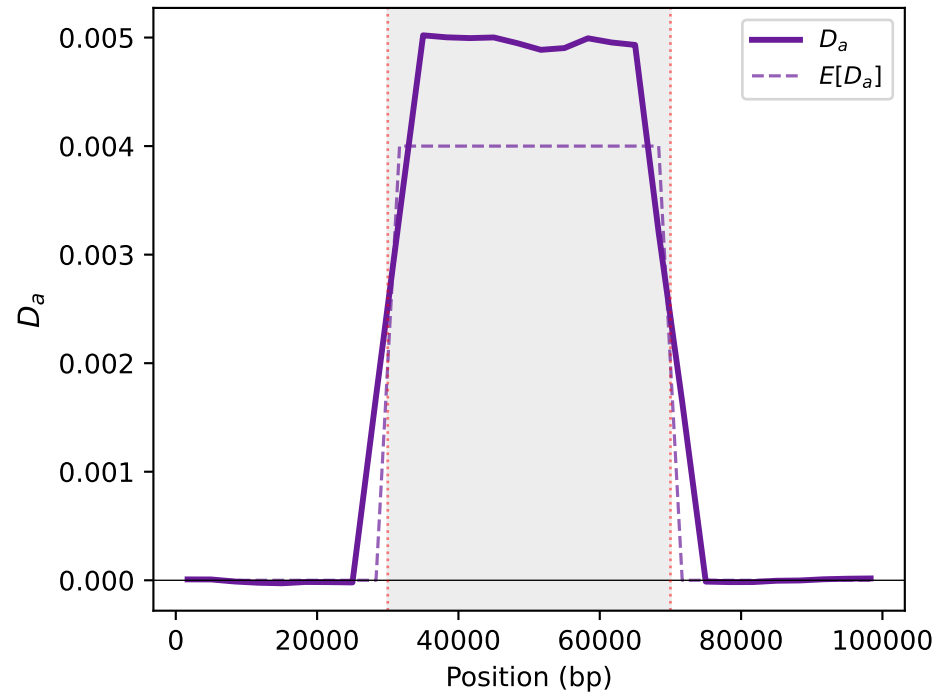
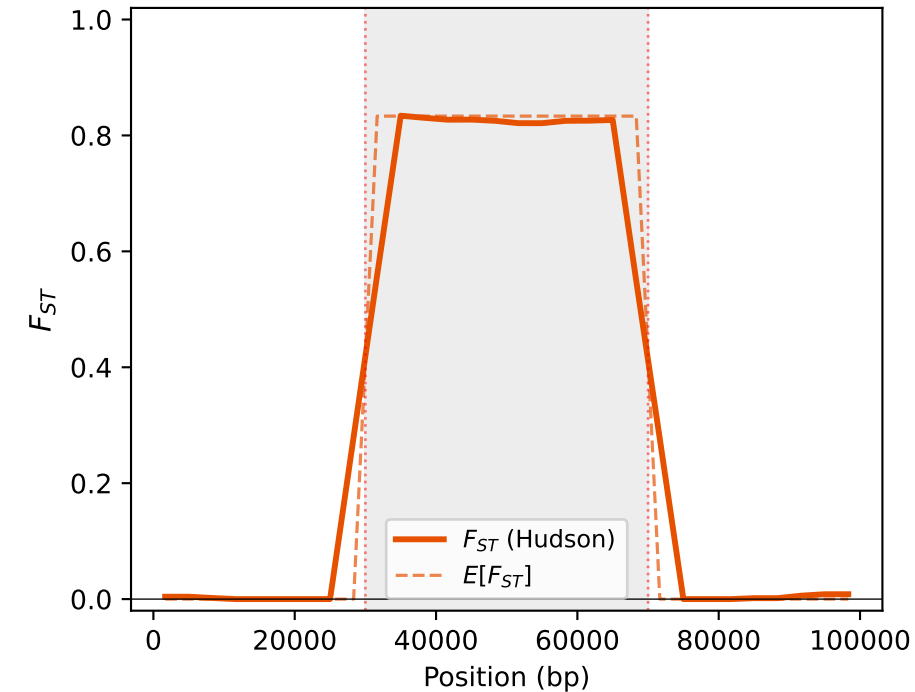
B. Net divergence $D_a = d_{XY} - (\pi_S + \pi_I)/2$ C. Hudson F_{ST} (relative differentiation)

Figure 1. Chromosomal inversion divergence signal simulated with msinv (Rust ARG core). (A) Per-bp absolute divergence (d_{XY}) between Standard (S) and Inverted (I) karyotypes is elevated inside the inversion (grey shading, 30–70 kb), while within-class diversity (π_S , π_I) remains at background levels. (B) Net divergence $D_a = d_{XY} - (\pi_S + \pi_I)/2$ isolates the barrier signal. (C) Hudson $F_{ST} = 1 - \pi_W/d_{XY}$ shows relative differentiation, clipped to [0, 1]. Parameters: $N_e=50,000$, $t_{inv}=200,000$ gen, $\rho=200$, $\gamma=1e-9$, $L=100$ kb, $\mu=1e-8$, $n_S=n_I=8$, 100 replicates.